

The Value of Delayed Information: Predictive-Information Envelopes for Age-Limited Control

Exact Bayes envelope generally; mutual-information bound universally;
closed form in the symmetric Markov case

Andrew H. Bond, *Senior Member, IEEE*
Department of Computer Engineering, San José State University
andrew.bond@sjsu.edu | ORCID: 0009-0003-2599-6158

Working draft — v0.1 (theory core complete; battery + related work open)

Abstract

A controller that acts on information of age D — a sensing delay, a telemetry lag, a stale cache — can outperform the best static policy only to the extent that the delayed observation still *predicts* the current state. Our companion systems work [1] proves this exactly for symmetric two-state Markov capacity ($\Delta(D) = \frac{1}{2}r(D)$, the autocorrelation at the sensing lag) with a β -mixing converse. This paper supplies the general theory. We (i) isolate the exact object — the *Bayes envelope* $\Delta_Y = \mathbb{E}[\psi(\mu_Y)] - \psi(\bar{\mu})$, the expected excess value of posterior over prior beliefs; (ii) prove the universal *predictive-information envelope* $0 \leq \Delta_Y \leq \Omega\sqrt{\frac{1}{2}I(X;Y)}$ for any bounded objective with oscillation Ω , by Pinsker’s inequality applied to the posterior process; (iii) derive the *bit-budget corollary*: a telemetry path delivering b bits per decision caps feedback value at $\Omega\sqrt{(b\ln 2)/2}$ regardless of what is encoded — compression can preserve but never create feedback value; (iv) show the envelope is *tight to leading order* in the canonical case: for the symmetric binary chain $I = \frac{r^2}{2} + O(r^4)$, so the bound reads $\Omega(\frac{r}{2} + O(r^3))$ against the exact $\frac{1}{2}r$; and (v) prove by a two-bit construction that no universal equality $\Delta = f(I)$ can hold: mutual information is a scalar, decision value is a geometry, and two observations with identical I can carry all or none of the decision-relevant distinction. A five-domain empirical section (cluster contention, acoustic, seismic, space weather, bioacoustic) tests the law and its envelopes on real processes.

1 Introduction

[TODO: motivation prose; position vs. AoI (freshness metrics without decision semantics), value-of-information (Howard [2]; De Lara-Gossner [3]), remote estimation, predictive information (Bialek et al.), delayed-sharing structures (Witsenhausen). The one-line thesis: *age matters only through predictive information, and predictive information matters only through the Bayes envelope.*]

2 Setting and the exact object

Let X be the current exogenous state taking values in a finite set $\mathcal{X}$, with marginal $\bar{\mu}$. A controller observes a delayed quantity Y (a delayed sample X_{t-D} , a delayed history, or any $\sigma(\mathcal{F}_{\leq t-D})$ -measurable statistic), chooses an action w , and receives a bounded one-step reward $R(w, x)$ with

oscillation $\Omega = \max_{w,x} R - \min_{w,x} R$. Define the value of a belief $\mu \in \Delta(\mathcal{X})$:

$$\psi(\mu) = \max_w \sum_x R(w, x) \mu(x),$$

a pointwise maximum of linear functionals, hence convex; and the posterior $\mu_Y = \mathbb{P}(X \in \cdot \mid Y)$, with $\mathbb{E}[\mu_Y] = \bar{\mu}$ by the tower property. The optimal Y -measurable controller attains $V_Y^* = \mathbb{E}[\psi(\mu_Y)]$; the best static controller attains $V_{\text{stat}} = \psi(\bar{\mu})$. The exact object of this paper is the **Bayes envelope**

$$\Delta_Y = V_Y^* - V_{\text{stat}} = \mathbb{E}[\psi(\mu_Y)] - \psi(\bar{\mu}) \geq 0, \quad (1)$$

nonnegative by Jensen. Everything that follows is an envelope on (1); the companion paper's exact law is the case where (1) admits a closed form.

Lemma 1 (Lipschitz value [1, App. A]). $|\psi(\mu) - \psi(\mu')| \leq \Omega \|\mu - \mu'\|_{\text{TV}}$, hence $\Delta_Y \leq \Omega \mathbb{E} \|\mu_Y - \bar{\mu}\|_{\text{TV}}$.

The total-variation route specializes to the β -mixing converse of [1] when Y is the full delayed history: $\Delta(D) \leq \Omega \beta(D)$.

3 The predictive-information envelope

Theorem 2 (Predictive-information bound). *For any bounded objective with oscillation Ω and any delayed observable Y ,*

$$0 \leq \Delta_Y \leq \Omega \sqrt{\frac{1}{2} I(X; Y)} \quad (I \text{ in nats; in bits, } \Omega \sqrt{\frac{\ln 2}{2} I_{\text{bits}}}). \quad (2)$$

Proof. By Lemma 1, $\Delta_Y \leq \Omega \mathbb{E} \|\mu_Y - \bar{\mu}\|_{\text{TV}}$. Pinsker's inequality gives, for every realization y , $\|\mu_y - \bar{\mu}\|_{\text{TV}} \leq \sqrt{\frac{1}{2} D_{\text{KL}}(\mu_y \parallel \bar{\mu})}$. Taking the expectation over Y and applying Jensen to the concave square root,

$$\mathbb{E} \|\mu_Y - \bar{\mu}\|_{\text{TV}} \leq \mathbb{E} \sqrt{\frac{1}{2} D_{\text{KL}}(\mu_Y \parallel \bar{\mu})} \leq \sqrt{\frac{1}{2} \mathbb{E} D_{\text{KL}}(\mu_Y \parallel \bar{\mu})} = \sqrt{\frac{1}{2} I(X; Y)},$$

the last equality being the definition of mutual information as the expected KL divergence of posterior from prior. $\square$

Corollary 3 (Bit-limited telemetry). *If the sensing path delivers at most b bits per decision — $Y = g(Z)$ with Z the raw delayed signal and $H(Y) \leq b \ln 2$ nats — then $\Delta_Y \leq \Omega \sqrt{(b \ln 2)/2}$, and moreover $\Delta_Y \leq \Omega \sqrt{\frac{1}{2} I(X; Z)}$ regardless of g (data processing). Compression can preserve feedback value; no encoding can create it.*

Proposition 4 (Leading-order tightness in the canonical case). *For the symmetric two-state Markov chain with lag- D autocorrelation $r = r(D)$,*

$$I(X_t; X_{t-D}) = \frac{1}{2} [(1+r) \ln(1+r) + (1-r) \ln(1-r)] = \frac{r^2}{2} + \frac{r^4}{12} + O(r^6),$$

so the right side of (2) equals $\Omega(\frac{r}{2} + O(r^3))$, matching the exact law $\Delta(D) = \frac{1}{2} r(D)$ of [1] to leading order (at the objective normalization $\Omega = 1$). The information envelope is not loose where it matters.

Proof. The joint distribution places mass $\frac{1+r}{4}$ on each agreeing pair and $\frac{1-r}{4}$ on each disagreeing pair; the stated I follows by direct computation, and the expansion from $(1 \pm r) \ln(1 \pm r) = \pm r + \frac{r^2}{2} \mp \frac{r^3}{6} + \dots$. Then $\sqrt{\frac{1}{2} I} = \frac{r}{2} \sqrt{1 + \frac{r^2}{6} + O(r^4)}$. $\square$

4 No universal equality $\Delta = f(I)$

Proposition 5 (Information is a scalar; value is a geometry). *There is a state space, a bounded objective, and two observations Y_1, Y_2 with $I(X; Y_1) = I(X; Y_2) > 0$ but $\Delta_{Y_1} = \frac{\Omega}{2} \cdot \frac{1}{2} > 0 = \Delta_{Y_2}$. Hence no function f with $\Delta_Y = f(I(X; Y))$ exists for all experiments.*

Proof. Let $X = (B_1, B_2)$ be two independent fair bits and let the objective depend only on B_1 : actions $w \in \{0, 1\}$, $R(w, x) = \Omega \mathbf{1}[w = B_1]$ (so the oscillation is Ω). Static value: $\Omega/2$. Let $Y_1 = B_1$ and $Y_2 = B_2$. Both have $I(X; Y_i) = \ln 2$. Observing Y_1 yields $V^* = \Omega$, so $\Delta_{Y_1} = \Omega/2$; observing Y_2 leaves the posterior over B_1 uniform, so $\Delta_{Y_2} = 0$. The two experiments share every information quantity in $I(X; Y)$ yet differ maximally in value: the envelope (2) is attained within a constant on one and vacuous on the other. $\square$

Remark 6. *This is the precise sense in which the companion paper’s binary law is special: there the state is one decision-relevant bit, so the geometry collapses to a one-parameter family and the envelope inverts to an exact closed form. The general exact object remains (1).*

5 Envelope comparison

The three envelopes are not interchangeable, and their relationship is an ordered pair of relaxations of one master quantity.

Proposition 7 (Master envelope and universal ordering). *For any delayed observable Y measurable w.r.t. the delayed history $\mathcal{F}_{\leq t-D}$,*

$$\Delta_Y \leq \Omega \mathbb{E} \|\mu_Y - \bar{\mu}\|_{\text{TV}} \leq \Omega \min \left\{ \beta(D), \sqrt{\frac{1}{2} I(X; Y)} \right\},$$

where $\beta(D)$ is the β -mixing coefficient of the state process at lag D . The expected posterior movement $\mathbb{E} \|\mu_Y - \bar{\mu}\|_{\text{TV}}$ is the master envelope; the mixing and information envelopes are its two relaxations.

Proof. The first inequality is Lemma 1. For the second: since Y is $\mathcal{F}_{\leq t-D}$ -measurable and $\sigma(X) \subseteq \mathcal{F}_{\geq t}$, conditioning on the coarser Y can only shrink expected posterior movement, and $\mathbb{E} \|\mu_{\mathcal{F}_{\leq t-D}} - \bar{\mu}\|_{\text{TV}} \leq \beta(D)$ because the β -coefficient takes its supremum over the larger future field [1, App. A]. The information relaxation is the Pinsker step in the proof of Theorem 2. $\square$

Proposition 8 (The two relaxations are incomparable). *Neither relaxation dominates the other.*

- (i) *Mixing strictly tighter: for the symmetric two-state chain with $Y = X_{t-D}$, $\beta(D) = \frac{1}{2}r$ while $\sqrt{\frac{1}{2}I} = \frac{r}{2}\sqrt{1 + \frac{r^2}{6}} + O(r^4) > \beta(D)$ for every $r \in (0, 1)$.*
- (ii) *Information arbitrarily tighter: for every $K \geq 2$ there is a stationary state process and objective with $\beta(D) = \frac{1}{2}$ for all D (the mixing envelope never decays) while $\sqrt{\frac{1}{2}I(X; \mathcal{F}_{\leq t-D})} = \sqrt{\ln 2 / (2K)} \rightarrow 0$, and indeed $\Delta = \Omega / (2K)$.*

Proof. (i) is Proposition 4 together with $\beta(D) = \frac{1}{2}r(D)$ for the binary chain [1, Cor. binary]. (ii) *The preview process.* Let c_t be i.i.d. fair bits, let $\Theta \sim \text{Unif}\{0, \dots, K-1\}$ be an independent phase, and let the observable state be $X_t = (c_t, p_t)$ with

$$p_t = \begin{cases} c_{t+m} & t \equiv \Theta \pmod{K} \\ \perp & \text{otherwise,} \end{cases} \quad m = D + K,$$

a sensor that occasionally publishes a *forecast* m steps ahead. The objective depends only on the current payoff bit: $R(w, x) = \Omega \mathbf{1}[w = c]$. *Mixing*: the window $[t - m, t - D]$ has length K , hence contains exactly one preview slot $s \equiv \Theta$; that preview reveals $c_{s+m} \in \{c_t, \dots, c_{t+K}\}$, a present-or-future payoff bit, so some event in $\mathcal{F}_{\geq t}$ has conditional probability in $\{0, 1\}$ against marginal $\frac{1}{2}$: $\beta(D) = \frac{1}{2}$ for every D . *Information*: the delayed history determines c_t exactly when $t - m \equiv \Theta \pmod{K}$ (probability $1/K$ over the phase) and is independent of c_t otherwise, so $I(c_t; \mathcal{F}_{\leq t-D}) = \frac{\ln 2}{K}$. *Value*: feedback wins $\Omega/2$ on the revealed fraction $1/K$ and nothing elsewhere: $\Delta = \Omega/(2K)$. The mixing envelope is vacuous; the information envelope $\Omega \sqrt{\ln 2/(2K)}$ decays with K (at rate $K^{-1/2}$ against the true K^{-1} — loose, but not blind). $\square$

Remark 9 (Why keep the information form at all). *Proposition 7 shows the TV master envelope is always at least as tight, and (i) shows mixing can beat information. Three properties are the information envelope’s own: it is estimable — the plug-in $\hat{I}(X_t; X_{t-D})$ in §7 needs only pair counts, while the β -coefficient’s supremum over the future σ -field is not estimable without a process model; it composes with channels — data processing gives Corollary 3, and no analogous statement exists for β (mixing has no channel semantics); and it adds across sensors via the chain rule, giving budget statements for multi-sensor telemetry. In short: mixing is the sharper analysis tool when a model is in hand; information is the operational one when only data or a bit budget is.*

Envelope	Bound on Δ_Y	Needs	Blind spot
master (TV)	$\Omega \mathbb{E} \ \mu_Y - \bar{\mu}\ _{\text{TV}}$	posterior process	not estimable directly
mixing	$\Omega \beta(D)$	process model	future events beyond X_t (Prop. 8ii)
information	$\Omega \sqrt{I(X; Y)/2}$	pair samples	decision geometry (Prop. 5)
bit budget	$\Omega \sqrt{(b \ln 2)/2}$	channel rate only	everything but the rate

Table 1: The envelope family. Each successive row needs less knowledge and pays for it in tightness; the battery of §7 measures the first three on real processes.

6 Exactness beyond two states

[TODO: layer-cake lift (cite [1, Prop. 2]): exact $\Delta(D) = \sum_u \pi_u r_u(D)$ for monotone-threshold objectives — already a multi-state exact law; characterize the objective class for which the Bayes envelope admits closed form; asymmetric two-state case.]

7 The law and its envelopes across domains

The companion paper had room for one paragraph of this study; the full protocol and results are reported here. The question has two halves. First, *the law*: is $\Delta(D) = \frac{1}{2} r(D)$ — exact for symmetric binary Markov capacity — also the operative value of feedback on *real* binary state processes, i.e., does history beyond the single delayed sample add value? Second, *the envelope*: does the measured value respect, and how tightly does it approach, the predictive-information bound of Theorem 2?

7.1 Protocol

States. Binary regime series from four physical domains plus controls: acoustic source azimuth and voice activity (LOCATA, 120 Hz); sperm-whale coda type (Dominica corpus, per-exchange);

GOES space-weather series (X-ray flux, magnetometer, proton flux; 1–5-minute cadence); seismic station regime (ANMO, 1-minute); a synthetic two-state Markov chain (flip probability 0.02) as positive reference; an i.i.d. fair-bit series as null. Each series is binarized by the domain’s registered thresholding (the companion study’s preparation scripts, unchanged).

The two estimands. For each lag D on a per-domain grid: (i) the pooled lag- D autocorrelation $r(D)$, giving the law value $\frac{1}{2}r(D)$; (ii) the value of the best *delayed-history* policy $\hat{\Delta}_{\text{hist}}(D)$: a $k=6$ -lag lookup predictor trained and evaluated by blocked cross-validation (5 folds, contiguous blocks, no leakage across fold boundaries), reported as accuracy above chance — the operational value of feedback for the 0/1 prediction objective ($\Omega = 1$, for which the binary law reads exactly $\frac{1}{2}r$).

Surrogate calibration. The null for “history adds nothing beyond the law” is a population of 60 Markov surrogates per domain, each matched to the real series’ lag-1 transition probabilities and evaluated by the identical pipeline. The gain $\hat{\Delta}_{\text{hist}} - \frac{1}{2}r$ is called significant only if its maximum over the lag grid exceeds the surrogate maxima at $p < 0.05$ and clears a practical floor of 0.03 (twice the i.i.d. null’s estimator fluctuation) — the dual criterion of the companion study.

The envelope measurement. New here: for each domain and lag, the plug-in mutual information $\hat{I}(X_t; X_{t-D})$ from pooled pair counts (2×2 table; plug-in bias $\leq 3/(2N)$ nats at the smallest pooled N in the battery, negligible at the reported scale), and the envelope $\sqrt{\hat{I}/2}$ of Theorem 2. Because $\hat{\Delta}_{\text{hist}}$ is a noisy estimate, the registered check is

$$\hat{\Delta}_{\text{hist}}(D) \leq \sqrt{\hat{I}(D)/2} + \varepsilon_{\text{est}},$$

with ε_{est} the maximum absolute fluctuation of the same estimator on the i.i.d. null — the null calibrates the noise floor, so an apparent “violation” smaller than what pure noise produces on a zero-information series is reported as estimator noise, not physics.

7.2 Results

Table 2 reports, per domain: the maximum history gain beyond the law with its surrogate p -value; the per-domain noise bound ε_{dom} (the maximum of the surrogates’ 97.5% gain band — surrogates have zero true history gain by construction and share the domain’s series lengths, so they calibrate its estimator noise); the maximum excess of the history policy over the *single-lag* envelope $\sqrt{\hat{I}(X_t; X_{t-D})/2}$; and the tightness ratio envelope/law (median over lags with $r > 0.05$).

Three readings, in increasing order of consequence.

The law, confirmed where it should hold. On the Markov reference and the three domains whose surrogate test finds nothing beyond the law (azimuth, whale codas, protons), the best delayed-history policy sits on $\frac{1}{2}r(D)$ across the lag grid — the single delayed sample is the whole value of feedback, in acoustics, bioacoustics, and space weather alike.

Tightness, observed in the wild. The envelope/law column is Proposition 4 measured: 1.001 on the reference chain (small r : the envelope sits *on* the law), rising to 1.099 on azimuth exactly as the $\sqrt{1+r^2/6}$ correction predicts for its slowly-decaying (large- r) regime. The information envelope is not a loose theoretical courtesy; on Markov-regime data it *is* the law, to within a predicted $O(r^2)$ factor.

The envelope as a non-Markovianity witness. In the lower block the history policy exceeds the single-lag envelope — decisively on VAD (+0.128 against a noise bound of 0.059) and seismic regime (+0.157 vs. 0.037). This is not a counterexample to Theorem 2: the history policy’s observable is the 6-lag window Y , and the theorem bounds it by $\sqrt{I(X_t; Y)/2}$ for *that* Y . Read

Domain	gain _{max}	p	ε_{dom}	$\max(\hat{\Delta} - \text{env}_1)$	env/law	verdict
Two-state Markov (ref)	0.030	.38	0.069	+0.023	1.001	law exact
i.i.d. null (control)	0.012	.68	0.016	+0.012	—	$\Delta \approx 0$
LOCATA azimuth	0.008	.05	0.007	−0.013	1.099	law exact
Sperm-whale codas	0.046	.98	0.112	+0.045	1.010	law exact
GOES protons	0.029	.77	0.048	+0.030	0.998	law exact
LOCATA VAD	0.100	.00	0.059	+0.128	0.940	history adds
GOES X-ray flux	0.185	.00	0.100	+0.071	0.937	history adds
GOES magnetometer	0.147	.00	0.070	+0.041	0.972	history adds
Seismic ANMO	0.367	.00	0.037	+0.157	0.999	history adds

Table 2: Battery v3 (nine series, four physical domains + controls; `battery_v3.json`). Upper block: the law is the whole value of feedback and the single-lag envelope is respected within noise. Lower block: the surrogate test certifies history value beyond the law ($p < .05$ and gain above the 0.03 floor), and exactly there the history policy can exceed the *single-sample* envelope — bold where the excess clears ε_{dom} , certifying window information beyond the single lag (see text).

backwards, the theorem converts the measured exceedance into a *certified information lower bound*:

$$I(X_t; \text{window}) \geq 2 \hat{\Delta}_{\text{hist}}^2,$$

which on the seismic series certifies ≥ 0.072 nats of window information at a lag where the single sample carries 0.002 — a $30\times$ concentration of predictive information in history beyond the last sample (aftershock clustering, in information units). The two blocks of Table 2 are therefore not “pass” and “fail”: they are the theorem operating in both directions — as an envelope on Markov-regime processes, and as a quantitative witness of history structure on non-Markov ones. The dual use is registered in the companion study’s terms: the same surrogate machinery that certifies the law’s exactness upper-block certifies the window-information excess lower-block.

8 Related work

[TODO: Howard VoI [2]; De Lara–Gossner [3]; AoI surveys; remote estimation / Witsenhausen delayed sharing; Bialek predictive information; rate-distortion with side information; NCS.]

9 Conclusion

[TODO:]

References

- [1] A. H. Bond. Polite bursting: the value of delayed feedback in age-limited admission control. Under submission, IEEE INFOCOM 2027.
- [2] R. A. Howard. Information value theory. *IEEE Trans. Systems Science and Cybernetics* 2(1):22–26, 1966.
- [3] M. De Lara, O. Gossner. Payoffs-beliefs duality and the value of information. *SIAM J. Optimization* 30(1):287–310, 2020.